Table 1: Topic-specific Datasets D_i

Name	Topic; Number of statements	Example; T/F = True/False
cities	Locations of cities; 1496	The city of Bhopal is in India. (T)
sp_en_trans	Spanish to English translations; 354	The Spanish word 'uno' means 'one'. (T)
element_symb	Symbols of elements; 186	Indium has the symbol As. (F)
animal_class	Classes of animals; 164	The giant anteater is a fish. (F)
inventors	Home countries of inventors; 406	Galileo Galilei lived in Italy. (T)
facts	Diverse scientific facts; 561	The moon orbits around the Earth. (T)

more robust LLM lie detector. As discussed in Section 6, our work opens the door to several future research directions in the general quest to construct more transparent, honest and safe AI systems.

2 Datasets with true and false statements

To explore the internal truth representation of LLMs, we collected several publicly available, labelled datasets of true and false English statements from previous papers. We then further expanded these datasets to include negated statements, statements with more complex grammatical structures and German statements. Each dataset comprises hundreds of factual statements, labelled as either true or false. First, as detailed in Table 1, we collected six datasets of affirmative statements, each on a single topic. The `cities` and `sp_en_trans` datasets are from Marks and Tegmark [2023], while `element_symb`, `animal_class`, `inventors` and `facts` are subsets of the datasets compiled by Azaria and Mitchell [2023]. All datasets, with the exception of `facts`, consist of simple, uncontroversial and unambiguous statements. Each dataset (except `facts`) follows a consistent template. For example, the template of `cities` is "The city of <city name> is in <country name>.", whereas that of `sp_en_trans` is "The Spanish word <Spanish word> means <English word>." In contrast, `facts` is more diverse, containing statements of various forms and topics.

Following Levinstein and Herrmann [2024], each of the statements in the six datasets from Table 1 is negated by inserting the word "not". For instance, "The Spanish word 'dos' means 'enemy'." (False) turns into "The Spanish word 'dos' does not mean 'enemy'." (True). This results in six additional datasets of negated statements, denoted by the prefix "`neg_`". The datasets `neg_cities` and `neg_sp_en_trans` are from Marks and Tegmark [2023], `neg_facts` is from Levinstein and Herrmann [2024], and the remaining datasets were created by us.

Furthermore, we use the DeepL translator tool to translate the first 50 statements of each dataset in Table 1, as well as their negations, to German. The first author, a native German speaker, manually verified the translation accuracy. These datasets are denoted by the suffix `_de`, e.g. `cities_de` or `neg_facts_de`. Unless otherwise specified, when we mention affirmative and negated statements in the remainder of the paper, we refer to their English versions by default.

Additionally, for each of the six datasets in Table 1 we construct logical conjunctions ("and") and disjunctions ("or"), as done by Marks and Tegmark [2023]. For conjunctions, we combine two statements on the same topic using the template: "It is the case both that [statement 1] and that [statement 2].". Disjunctions were adapted to each dataset without a fixed template, for example: "It is the case either that the city of Malacca is in Malaysia or that it is in Vietnam.". We denote the datasets of logical conjunctions and disjunctions by the suffixes `_conj` and `_disj`, respectively. From now on, we refer to all these datasets as topic-specific datasets D_i.

In addition to the 36 topic-specific datasets, we employ two diverse datasets for testing: `common_claim_true_false` [Casper et al., 2023] and `counterfact_true_false` [Meng et al., 2022], modified by Marks and Tegmark [2023] to include only true and false statements. These datasets offer a wide variety of statements suitable for testing, though some are ambiguous, malformed, controversial, or potentially challenging for the model to understand [Marks and Tegmark, 2023]. Appendix A provides further information on these datasets, as well as on the logical conjunctions, disjunctions and German statements.

3 Supervised learning of the truth directions

As mentioned in the introduction, we learn the truth directions from the internal model activations. To clarify precisely how the activations vectors of each model are extracted, we first briefly explain parts of the transformer architecture [Vaswani, 2017, Elhage et al., 2021] underlying LLMs. The input text is first tokenized into a sequence of h tokens, which are then embedded into a high-dimensional space, forming the initial residual stream state $\mathbf{x}_0 \in \mathbb{R}^{h \times d}$, where d is the embedding dimension. This state is updated by L sequential transformer layers, each consisting of a multi-head attention mechanism and a multilayer perceptron. Each transformer layer l takes as input the residual stream activation $\mathbf{x}_{l-1}$ from the previous layer. The output of each transformer layer is added to the residual stream, producing the updated residual stream activation $\mathbf{x}_l$ for the current layer. The activation vector $\mathbf{a}_L \in \mathbb{R}^d$ over the final token of the residual stream state $\mathbf{x}_L \in \mathbb{R}^{h \times d}$ is decoded into the next token distribution.

Following Marks and Tegmark [2023], we feed the LLM one statement at a time and extract the residual stream activation vector $\mathbf{a}_l \in \mathbb{R}^d$ in a fixed layer l over the final token of the input statement. We choose the final token of the input statement because Marks and Tegmark [2023] showed via patching experiments that LLMs encode truth information about the statement above this token. The choice of layer depends on the LLM. For LLaMA3-8B we choose layer 12. This is justified by Figure 2, which shows that true and false statements have the largest separation in this layer, across several datasets.

Following this procedure, we extract an activation vector for each statement s_{ij} in the topic-specific dataset D_i and denote it by $\mathbf{a}_{ij} \in \mathbb{R}^d$, with d being the dimension of the residual stream at layer 12 ($d = 4096$ for LLaMA3-8B). Here, the index i represents a specific dataset, while j denotes an individual statement within each dataset. Computing the LLaMA3-8B activations for all statements (≈ 45000) in all datasets took less than two hours using a single Nvidia Quadro RTX 8000 (48 GB) GPU.

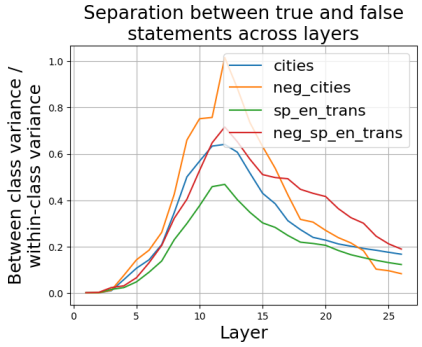

Figure 2: Ratio of the between-class variance and within-class variance of activations corresponding to true and false statements, across residual stream layers, averaged over all dimensions of the respective layer.

As mentioned in the introduction, we demonstrate the existence of *two* truth directions in the activation space: the general truth direction $\mathbf{t}_G$ and the polarity-sensitive truth direction $\mathbf{t}_P$. In Figure 1 we visualise the projections of the activations $\mathbf{a}_{ij}$ onto the 2D subspace spanned by our estimates of the vectors $\mathbf{t}_G$ and $\mathbf{t}_P$. In this visualization of the subspace, we choose the orthonormalized versions of $\mathbf{t}_G$ and $\mathbf{t}_P$ as its basis. We discuss the reasons for this choice of basis for the 2D subspace in Appendix B. The activations correspond to an equal number of affirmative and negated statements from all topic-specific datasets. The top left panel shows both the general truth direction $\mathbf{t}_G$ and the polarity-sensitive truth direction $\mathbf{t}_P$. $\mathbf{t}_G$ consistently points from false to true statements for both affirmative and negated statements and separates them well with an area under the receiver operating characteristic curve (AUROC) of 0.98 (bottom left panel). In contrast, $\mathbf{t}_P$ points from *false to true* for affirmative statements and from *true to false* for negated statements. In the top center panel, we visualise the affirmative truth direction $\mathbf{t}_A$, found by training a linear classifier solely on the activations of affirmative statements. The activations of true and false *affirmative* statements separate along $\mathbf{t}_A$ with a small overlap. However, this direction does not accurately separate true and false *negated* statements (top right panel). $\mathbf{t}_A$ is a linear combination of $\mathbf{t}_G$ and $\mathbf{t}_P$, explaining why it fails to generalize to negated statements.

Now we present a procedure for supervised learning of $\mathbf{t}_G$ and $\mathbf{t}_P$ from the activations of affirmative and negated statements. Each activation vector $\mathbf{a}_{ij}$ is associated with a binary truth label $\tau_{ij} \in \{-1, 1\}$ and a polarity $p_i \in \{-1, 1\}$.

$$\tau_{ij} = \begin{cases} -1 & \text{if the statement } s_{ij} \text{ is } \textit{false} \\ +1 & \text{if the statement } s_{ij} \text{ is } \textit{true} \end{cases} \tag{1}$$

$$p_i = \begin{cases} -1 & \text{if the dataset } D_i \text{ contains } \textit{negated} \text{ statements} \\ +1 & \text{if the dataset } D_i \text{ contains } \textit{affirmative} \text{ statements} \end{cases} \tag{2}$$

We approximate the activation vector $\mathbf{a}_{ij}$ of an affirmative or negated statement s_{ij} in the topic-specific dataset D_i by a vector $\hat{\mathbf{a}}_{ij}$ as follows:

$$\hat{\mathbf{a}}_{ij} = \boldsymbol{\mu}_i + \tau_{ij}\mathbf{t}_G + \tau_{ij}p_i\mathbf{t}_P. \tag{3}$$

Here, $\boldsymbol{\mu}_i \in \mathbb{R}^d$ represents the population mean of the activations which correspond to statements about topic i. We estimate $\boldsymbol{\mu}_i$ as:

$$\boldsymbol{\mu}_i = \frac{1}{n_i}\sum_{j=1}^{n_i}\mathbf{a}_{ij}, \tag{4}$$

where n_i is the number of statements in D_i. We learn $\mathbf{t}_G$ and $\mathbf{t}_P$ by minimizing the mean squared error between $\hat{\mathbf{a}}_{ij}$ and $\mathbf{a}_{ij}$, summing over all i and j

$$\sum_{i,j} L(\mathbf{a}_{ij}, \hat{\mathbf{a}}_{ij}) = \sum_{i,j} \|\mathbf{a}_{ij} - \hat{\mathbf{a}}_{ij}\|^2. \tag{5}$$

This optimization problem can be efficiently solved using ordinary least squares, yielding closed-form solutions for $\mathbf{t}_G$ and $\mathbf{t}_P$. To balance the influence of different topics, we include an equal number of statements from each topic-specific dataset in the training set.

Figure 3 shows how well true and false statements from different datasets separate along $\mathbf{t}_G$ and $\mathbf{t}_P$. We employ a leave-one-out approach, learning $\mathbf{t}_G$ and $\mathbf{t}_P$ using activations from all but one topic-specific dataset (including both affirmative and negated versions). The excluded datasets were used for testing. Separation was measured using the AUROC, averaged over 10 training runs on different random subsets of the training data. The results clearly show that $\mathbf{t}_G$ effectively separates both affirmative and negated true and false statements, with AUROC values close to one. In contrast, $\mathbf{t}_P$ behaves differently for affirmative and negated statements. It has AUROC values close to one for affirmative statements but close to zero for negated statements. This indicates that $\mathbf{t}_P$ separates affirmative and negated statements in reverse order. For comparison, we trained a Logistic Regression (LR) classifier with bias $b = 0$ on the centered activations $\tilde{\mathbf{a}}_{ij} = \mathbf{a}_{ij} - \boldsymbol{\mu}_i$. Its direction $\mathbf{d}_{LR}$ separates true and false statements similarly well as $\mathbf{t}_G$. We will address the challenge of finding a well-generalizing bias in Section 5.

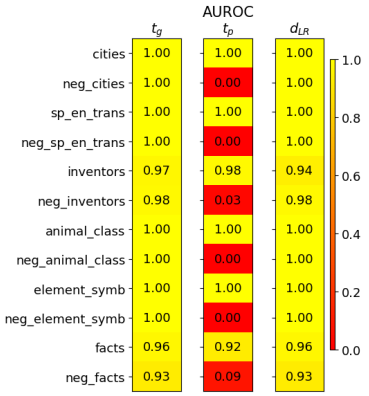

Figure 3: Separation of true and false statements along different truth directions as measured by the AUROC.

4 The dimensionality of truth

As discussed in the previous section, when training a linear classifier only on affirmative statements, a direction $\mathbf{t}_A$ is found which separates well true and false affirmative statements. We refer to $\mathbf{t}_A$ and the corresponding one-dimensional subspace as the affirmative truth direction. Expanding the scope to include negated statements reveals a *two*-dimensional truth subspace. Naturally, this raises questions about the potential for further linear structures and whether the dimensionality increases again with the inclusion of new statement types. To investigate this, we also consider logical conjunctions and disjunctions of statements, as well as statements that have been translated to German, and explore if additional linear structures are uncovered.

4.1 Number of significant principal components

To investigate the dimensionality of the truth subspace, we analyze the fraction of truth-related variance in the activations $\mathbf{a}_{ij}$ explained by the first principal components (PCs). We isolate truth-related variance through a two-step process: (1) We remove the differences arising from different sentence structures and topics by computing the centered activations $\tilde{\mathbf{a}}_{ij} = \mathbf{a}_{ij} - \boldsymbol{\mu}_i$ for all topic-specific datasets D_i; (2) We eliminate the part of the variance within each D_i that is uncorrelated